Joel Andrew Miller

(905) 933-7920 | joelmiller0430@gmail.com| joelandrewmiller.com

M.E.Sc. Machine Learning Engineer specializing in physiological signal processing and AI for human-machine systems. Experienced in building wearable sensor pipelines, biometric data analysis, and deploying ML models to interpret human behaviour in high-stakes environments.

WORK EXPERIENCE:

Machine Learning Engineer – Biological Signals | **ITPS Canada** | Feb 2026 – Present

- Develop and validate ML models to detect indicators of pilot stress and elevated cognitive workload from multimodal signals (ECG, Pupil Data, flight logs), enabling behavioural assessment in high-stakes flight environments.
- Designed and built a structured flight database to organize and query data by pilot, sortie type, and in-flight events, enabling systematic extraction of behavioural insights to inform human factors research and system design.

R&D Aerospace Software Engineer | **ITPS Canada** | Feb 2025 – Feb 2026

- Built a helmet-mounted display project from scratch, including an outside-in optical tracking pipeline, and co-designing in-house display.
- Designed all software for head tracking and sensor-fusion algorithms, achieving ~2° spatial accuracy in cockpit environments.
- Implemented a Kalman filter to correct IMU drift and maintain stable, long-duration tracking under high-motion and vibration conditions.
- Architected and implemented the distributed system enabling fully wireless operation suitable for aircraft with ejection seats.
- Engineered and optimized the UDP networking layer to ensure high-frequency, low-jitter pose transmission, achieving ~3 ms end-to-end latency.
- Designed and executed calibration workflows to align helmet, optical, and aircraft coordinate frames for consistent head-pose reconstruction.
- Collaborated directly with pilots, test engineers, program stakeholders, and leadership to translate operational needs into clear technical requirements and deliverables.

R&D Engineering Intern | **NRC Canada** | Sept 2023 – Dec 2023

- Researched and designed a pipeline to transform real-world sensor data (LiDAR, camera, GPS) collected from a vehicle to simulated environments for autonomous vehicle development and testing.

Software Engineer/DevOps Intern | **IBM** | May 2021 – Sept 2022

- Developed automated testing frameworks for IBM's Order Management System, reducing testing time by 90% and improving system reliability.
- Configured a Dockerized virtual machine integrated with a web application to execute and monitor automated test cases.
- Mentored and onboarded incoming interns, supporting team ramp-up and knowledge transfer.
- Built frontend components to improve user interaction and experience.

EDUCATION:

M.E.Sc Software Engineering with Specialization in Artificial Intelligence | **Western University** | May 2023 – Dec 2024

- Employed signal processing, data fusion, time series analysis, and deep learning to model and understand driver's cognitive behaviour through their physiological signals (ECG, EDA, RSP).

B.E.Sc Software Engineering with Co-op | **Western University** | 2018 – 2023

PROJECTS and LEADERSHIP:

Peer Reviewed Studies

- **Published**: "**Navigating the Handover: Reviewing Takeover Requests in Level 3 Autonomous Vehicles**" (IEEE Open Journal of Vehicular Technology): A comprehensive review of Takeover Requests (TORs) in autonomous vehicles, emphasizing safe human-centered transitions from automation to manual control.
- **Published**: "**At the Heart of Intersections: Analyzing Their Influence on Driver Heart Behaviour**" (Springer Data Science for Transportation): Investigated intersections' impact on driver heart rates to inform safer urban traffic designs and autonomous vehicle technologies.
- **Published**: "**Modeling Driver State and Takeover Behavior in Conditional Automation Using Physiological Signals**" (IEEE): Applied ML to ECG, EDA, and RSP signals to classify takeover occurrence and timing with 76% and 75% accuracy with a small amount of data (>30 samples), advancing adaptive, human-centric autonomous vehicle systems.

CARLA Reinforcement Learning Agent (*Masters Project)*

- Designed and implemented a reinforcement learning agent for autonomous vehicle navigation in the CARLA simulation environment, leveraging Deep Q-Learning and PPO algorithms.
- Fine-tuned reward functions to balance efficiency and exploitation, improving learning rates and decision-making accuracy.

FoodSnap (*Capstone Project)*

- Engineered a machine learning application to analyze the nutritional content of meals from single images, integrating image segmentation, depth estimation networks, and the USDA API.
- Achieved accurate identification of food types and volume estimation, enabling precise calorie and macronutrient calculations.
- Demonstrated the project's scalability and potential impact in nutrition tracking through real-world testing scenarios.

Western AI Club (*Project Manager)*

- Led a team of five developers to design, train, and deploy an iOS image classification app for waste categorization, promoting sustainable disposal practices.
- Implemented transfer learning techniques to optimize the model, achieving 98% accuracy in classifying waste types.
- Oversaw project milestones, ensuring timely delivery and seamless integration of app features.

Skills: Python, PyTorch, Tensorflow, scikit-learn, pandas, NumPy, OpenCV, SQL, signal processing, ECG/EDA/RSP analysis, wearable sensors, sensor fusion, Kalman filtering, IMU processing, camera calibration, optical tracking, anomaly detection, machine learning, computer vision, reinforcement learning, real-time systems, UDP networking, distributed systems, Docker, Git, human-machine interfaces, Linux.